

RS004 Week 1 — English Worksheet

Topic: Game Window & Game Loop · **Course:** Platformer Game · **Time:** about 45 minutes

This week you open your very first **game window** and start the **game loop** — the heartbeat that runs every frame: read events → update → draw. Everything you build for the next 16 weeks lives inside this loop.

Keep these words handy: **game loop**, **event**, `pygame.QUIT`, `flip()`, **FPS**, **RGB**.

1 · Predict 🌟

Read each snippet. Before you run it, write what you think will happen.

```
screen.fill((135, 206, 235))
pygame.display.flip()
```

What colour fills the screen, and what does `flip()` do?


```
for event in pygame.event.get():
    if event.type == pygame.QUIT:
        running = False
```

A player clicks the X button. What happens to `running`, and then to the loop?


```
clock.tick(60)
```

This line is at the end of the loop. What is it controlling?

2 · Read & Match

Match each line to its job. Write the letter next to each number.

1. `pygame.init()`
2. `pygame.display.set_mode((800, 600))`
3. `pygame.display.set_caption("나의 플랫폼머")`
4. `screen.fill(SKY_BLUE)`
5. `pygame.display.flip()`

- A. paints the whole screen one colour
- B. starts up pygame so we can use it
- C. shows everything we drew this frame
- D. makes the 800×600 window
- E. writes the title on the window bar

1 → ___ 2 → ___ 3 → ___ 4 → ___ 5 → ___

3 · Spot the Bug 🐛

Each snippet is broken. Read what it should do, then rewrite it so it works. Explain why the original was wrong.

Bug A — The window should stay open until the player closes it. Right now it flashes open and shuts instantly.

```
running = True
for event in pygame.event.get():
    if event.type == pygame.QUIT:
        running = False
screen.fill((135, 206, 235))
pygame.display.flip()
```

Hint: the drawing and event-reading need to *repeat*. What kind of loop is missing?

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The screen should turn sky-blue, but the window stays black even though `fill` runs.

```
screen.fill((135, 206, 235))
clock.tick(60)
```

Hint: we filled the screen but never *showed* it.

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Colour Lab

Each colour is `(Red, Green, Blue)`, where each number is 0–255. Write down what colour you expect, then check it in code.

```
(255, 0, 0)    → _____  
(0, 0, 0)     → _____  
(255, 255, 255) → _____  
(135, 206, 235) → _____
```


Now invent one. Write an RGB value for a sunset orange sky, then test it in `screen.fill(...)`.

5 · Build & Show 📷

Open your starter file. You have a working 800×600 sky-blue window at 60 FPS. Make it yours:

1. Change the background to a colour you like, and write down what its RGB numbers mean.
2. Change the window size to 1024×768.
3. Give the window your own caption (title).

When it works, send a **photo or video** of your window running, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I changed the background to ...

The three numbers in the colour mean ...

I made the window bigger by ...

The game loop keeps the window open because ...

One thing I was unsure about was ...

If I had more time, I would ...

6 · Record Your Walkthrough 🎥

Take a video on your phone while your window runs. Talk through it like you are teaching someone who has never seen Pygame. Try to use these words: **game loop, event, fill, flip, FPS**.

1. Show your window opening.
2. Point at the line that sets the background colour.
3. Explain what the game loop does every frame.
4. Click the X and show the window closing — say which line made that work.

Write what you will say in your video. Plan it here before you record — you can read from it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.